

Spatial Autocorrelation I

HES 505 Fall 2025: Session 25

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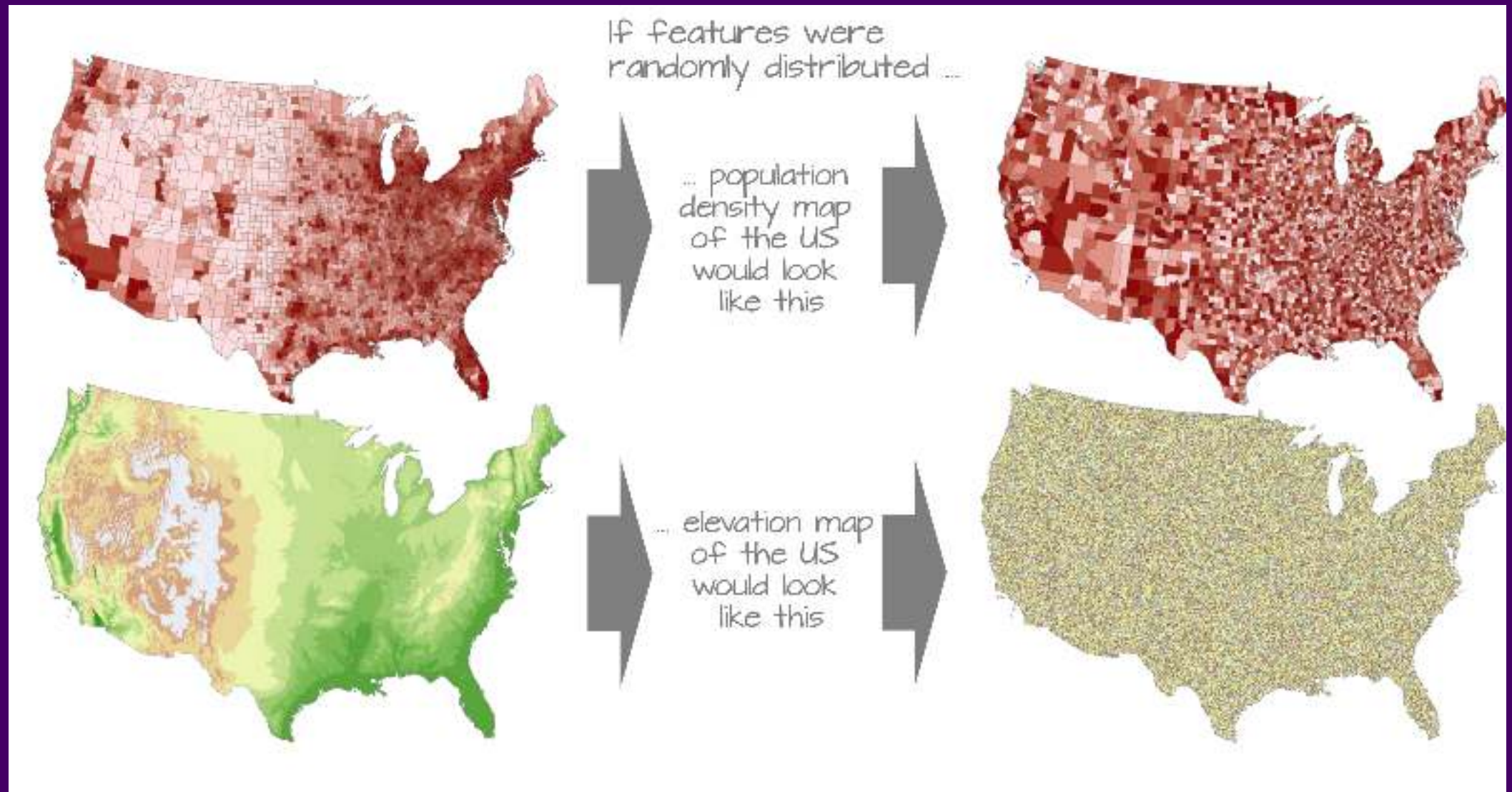
Objectives

By the end of today you should be able to:

- Use the **spdep** package to identify the neighbors of a given polygon based on proximity, distance, and minimum number
- Understand the underlying mechanics of Moran's I and calculate it for various neighbors
- Distinguish between global and local measures of spatial autocorrelation
- Visualize neighbors and clusters

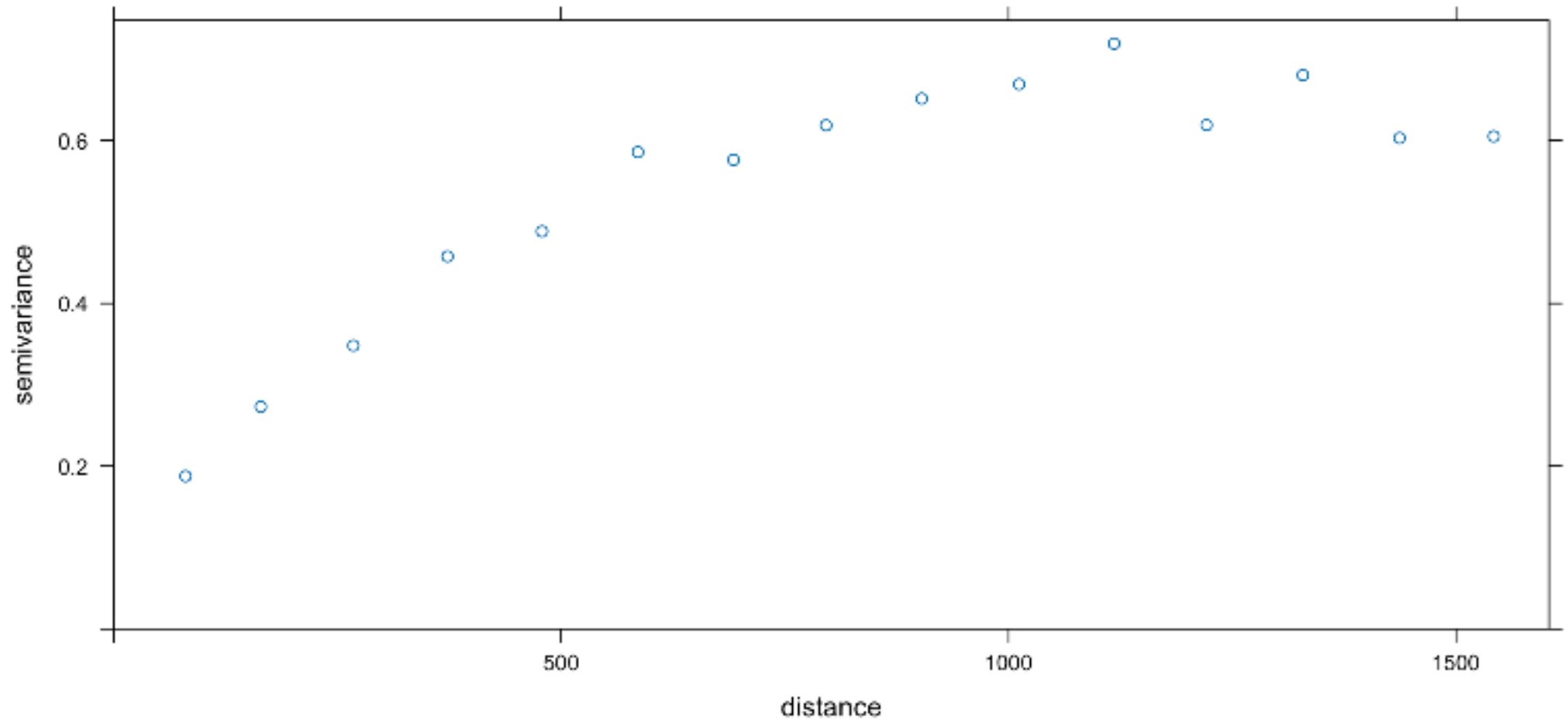
Revisiting Spatial Autocorrelation

The World Is Not Random



From Manuel Gimond

Kriging Revisited



Kriging Revisited

- Assumes autocorrelation to facilitate prediction
- Semivariogram describes how autocorrelation changes with distance
- But what about statistical inference?

Spatial Autocorrelation and Inference

- Attributes (features) are often non-randomly distributed
- Especially true with aggregated data
- Affects estimates of precision (and significance)

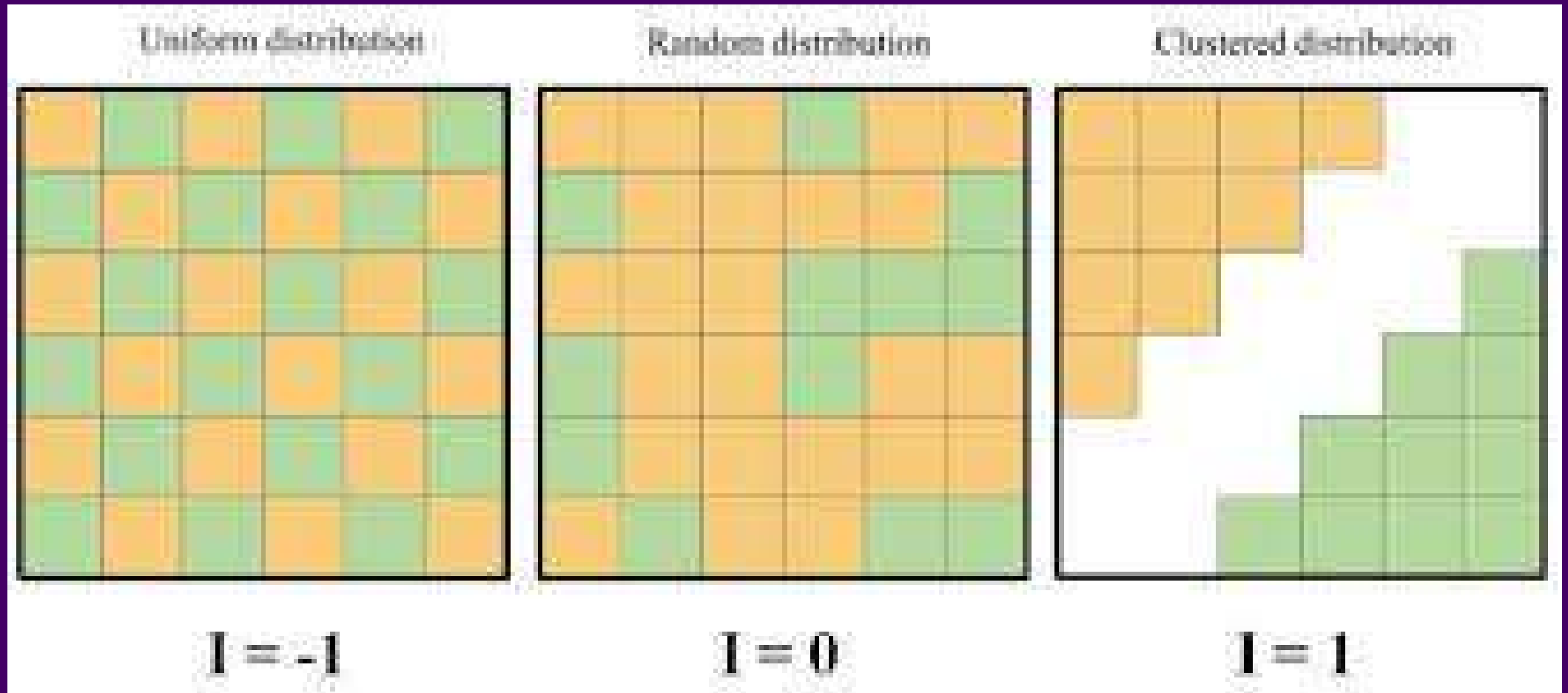
Global Moran's I

$$I = \frac{n}{W} \cdot \frac{\sum_{i=1}^n \sum_{j=1}^n w_{ij} (x_i - \bar{x})(x_j - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

where $W = \sum_{i=1}^n \sum_{j=1}^n w_{ij}$

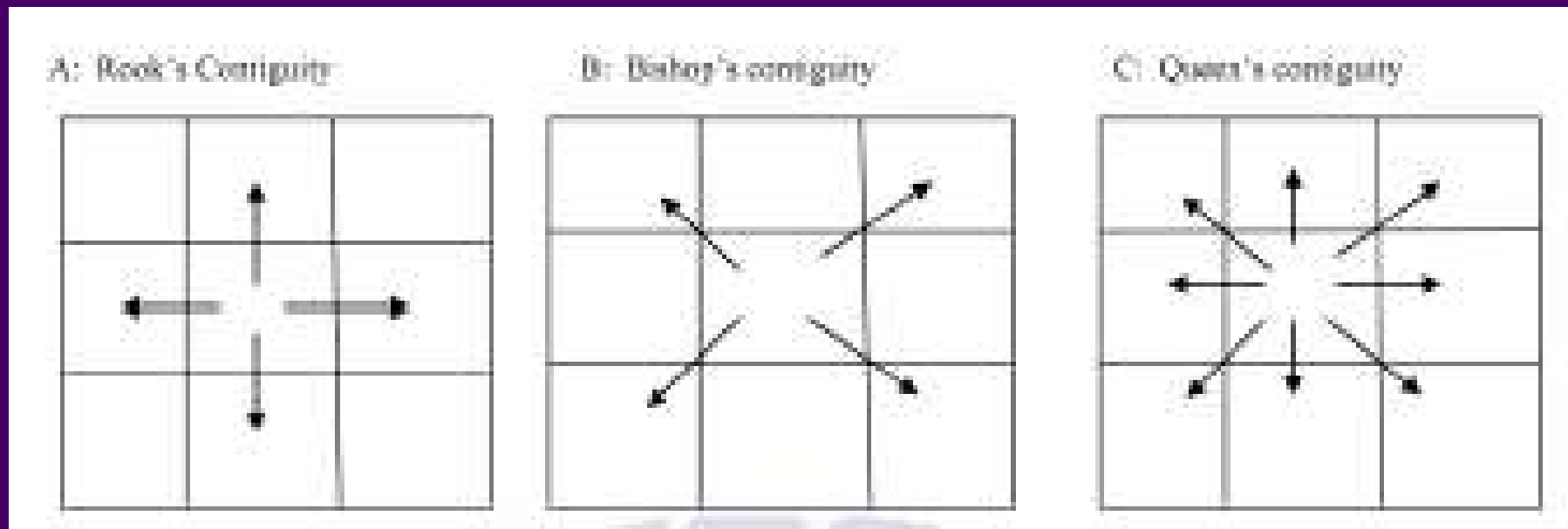
- A *first-order* estimate of spatial autocorrelation

Moran's I



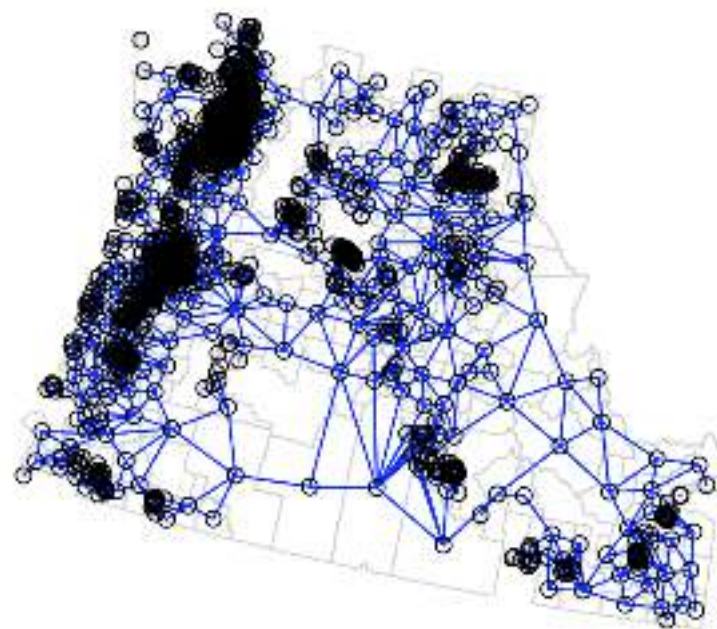
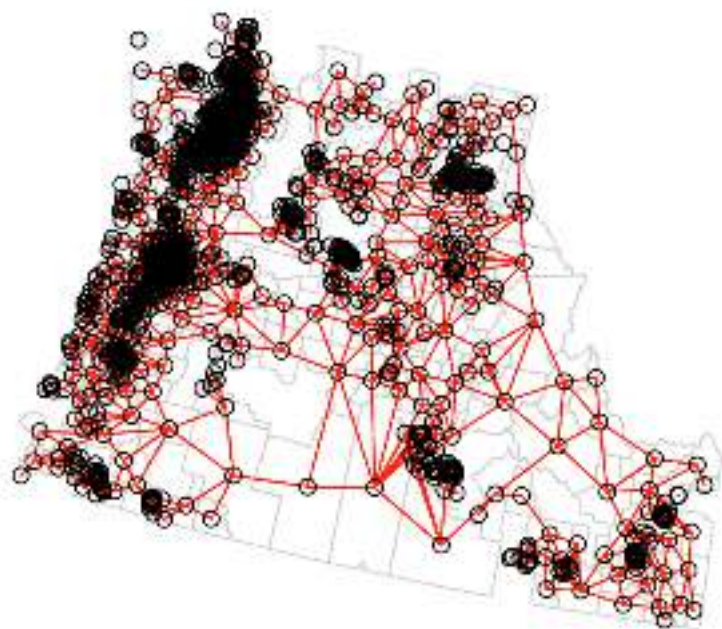
...but How Do We Calculate w_i

- How do we define $I(d)$ for areal data?
- What about w_{ij} ?
- We can use **spdep** for that!!



Finding Neighbors

- Queen, rook, (and bishop) cases impose neighbors by contiguity
- Weights calculated as a $1/\text{num. of neighbors}$



More Formally

$$E[I] = \frac{-1}{n-1}$$

- When n is large, $E[I]$ approaches 0
- But large relative to what?

Testing for Spatial Autocorrelation

- I can be estimated by fitting a model of the lagged average as a function of the measurement

$$\mu_{\text{lag}} = \beta \times \text{measurement}$$

where $\beta = I$

Comparing observed to expected

- We can generate the expected distribution of Moran's I coefficients under a Null hypothesis of no spatial autocorrelation
- Using permutation and a loop to generate simulations of Moran's I

Significance testing

- Pseudo p-value (based on permutations)
- Analytically (sensitive to deviations from assumptions)
- Using Monte Carlo

Local Indicators of Spatial Autocorrelation

- Sometimes we want to know about *second-order* autocorrelation
- Is there clustering around different locations (but maybe not everywhere)?

Local Moran's I

$$I_i = \frac{(x_i - \bar{x})}{\sum_{k=1}^n (x_k - \bar{x})^2 / n} \sum_{j=1}^n w_{ij} (x_j - \bar{x})$$

- Can identify areas where clustering is greater than expected
- Proportional to Global Moran's I

Autocorrelation and Inference

- Interpolation assumes autocorrelation
- Inference is interested in mechanisms
- Tests for mechanisms get confused by autocorrelation
- Diagnosing autocorrelation helps plan next steps (Weds)

